

KeePass 2.0

Cross-Platform Password Manager with Browser Integration

Forge Hackathon 2026 Submission

Field	Details
Project	KeePass 2.0 - Cross-Platform Password Manager
Hackathon	Forge Hackathon 2026
Date	August 11, 2026
Team	Shailesh Kolap, Vignesh Subramanian
Repository	github.com/Vigneshsub30/keepass2.0
Forge Project	SoftwareForge.ai
Demo Video	(Google Drive link - to be added)

Problem Statement

KeePass is a widely trusted, open-source password manager - but it has been locked to Windows and WinForms for over two decades. macOS users have had no native option, and there is no browser integration for auto-filling credentials. The codebase also had security gaps that needed addressing.

Our goal: Transform KeePass into a modern, cross-platform password manager with native macOS support and seamless browser integration - all in a single hackathon session, powered by AI-assisted development through Cursor and SoftwareForge.ai.

What We Built

Before (Original KeePass)

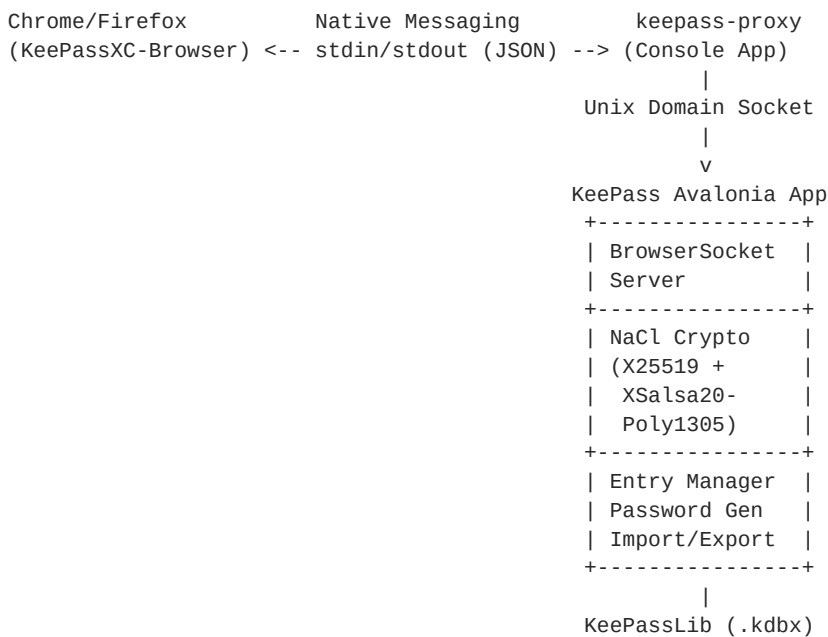
- Windows-only WinForms application
- No macOS or Linux GUI support
- No browser extension integration
- Security issues: global TLS bypass, unsigned plugins, no vault integrity checks

After (Our Hackathon Work)

- Cross-platform desktop app built with Avalonia UI (MVVM architecture), supporting macOS, Linux, and Windows (end-to-end tested on macOS ARM64)
- Browser extension integration via the KeePassXC-Browser protocol
- Full credential management: create, edit, delete, search entries
- Password generator with configurable strength options
- Import/Export support for KDBX and CSV formats
- Custom app icon and polished DMG installer

- Security hardening across the entire codebase

Architecture



Key Technical Achievements

1. Cross-Platform UI with Avalonia (MVVM)

Replaced the Windows-only WinForms frontend with a cross-platform Avalonia-based desktop app following the MVVM pattern. Supports macOS, Linux, and Windows from a single codebase. End-to-end tested on macOS ARM64.

2. Browser Extension Integration (KeePassXC-Browser Protocol)

Implemented the full KeePassXC-Browser native messaging protocol, enabling the existing Chrome/Firefox extension to communicate with our app for credential auto-fill.

3. End-to-End Encrypted Communication (NaCl)

All browser-to-app communication is encrypted using NaCl crypto_box (X25519 key exchange + XSalsa20-Poly1305 authenticated encryption) via a native messaging proxy.

4. Full Credential Lifecycle

Complete CRUD operations for password entries with a configurable password generator and import/export support for KDBX and CSV formats.

5. Security Hardening

Removed global TLS certificate validation bypass, added plugin signature verification, implemented vault integrity checks, and added security review sign-off gates.

Development Stats

Metric	Value
Commits	30
Files changed	62 (C#/XAML/project files)
Lines of code added	~5,000
Lines of code modified	~160
Forge work orders completed	15
Architecture Decision Records	8 ADRs
Development time	Single day
AI-assisted development	Cursor + SoftwareForge.ai

Forge Journey

This project was managed entirely through SoftwareForge.ai, which orchestrated the development workflow:

- Work Order Generation - Forge analyzed the KeePass codebase and generated targeted work orders covering security fixes, architectural decisions, and feature development.
- ADR-Driven Architecture - 8 Architecture Decision Records were produced to document key design choices (KDBX format selection, plugin trust model, clipboard delivery, MVVM adoption).
- Security-First Approach - Forge prioritized security work orders: TLS fix, plugin verification, vault integrity, and release promotion gates.
- Feature Development - After the foundation was solid, Forge guided feature work: cross-platform UI, browser integration, entry management, and packaging.
- Continuous Validation - Each work order included acceptance criteria verified through testing and validation.

Technology Stack

Layer	Technology
Language	C# / .NET 8
Desktop UI	Avalonia UI (MVVM, cross-platform)
Database	KeePassLib (KDBX 4.x format)
Crypto	NaCl.Net (X25519 + XSalsa20-Poly1305)
Browser IPC	Native Messaging Protocol (stdin/stdout)
App IPC	Unix Domain Sockets
Packaging	DMG (macOS), self-contained publish
AI Tools	Cursor IDE, SoftwareForge.ai

How to Run

macOS (DMG)

1. Download KeePass-2.61.1-osx-arm64.dmg from the artifacts
2. Double-click to mount the disk image
3. Drag KeePass Password Safe to your Applications folder
4. Launch from Applications (right-click > Open on first launch)
5. Open or create a .kdbx database file

Browser Extension

1. Install the KeePassXC-Browser extension for Chrome
2. With the app running and a database open, click the extension icon
3. Click Connect to pair the extension with the app
4. Navigate to any login page for auto-fill

Commit Log

Commit	Description
9fef109	Add app icon support in DMG packaging
4f0edfb	Update tests, solution and DMG packaging for new services
dc1e6b6	Wire up desktop UI: entry editor, preview panel, import/export, browser, quit
b0d2d4e	Add native messaging proxy for browser extension communication
cd16f90	Add core services: import/export, entry management, preview, browser protocol
122144d	Add artifacts/ to .gitignore
48443b0	Fix DMG creation for large single-file binaries
7cd2036	Fix DMG build: skip code signing when no certificate available
5680c48	Enable dual-publish: Avalonia on macOS/Linux, WinForms on Windows
ac4dc83	WO-105: Security review sign-off gate for release promotion
adfea5c	WO-103: Post-commit vault integrity check with rollback runbooks
cd30be3	WO-101: Add packaging scripts and fix Build/ path casing
6b6b219	WO-100: ADR for platform capability-tier matrix definition
3fa4233	WO-099: ADR for atomic-save transaction guarantees
90a21a7	WO-098: ADR for clipboard credential delivery mechanism
f08e74e	WO-097: ADR for UI-domain seam and MVVM adoption
6128d74	WO-096: ADR for Mono workaround retirement strategy
f5235cb	WO-095: ADR for configuration enforcement precedence hierarchy
f4b8f41	WO-094: ADR for plugin trust model and isolation
42d66ba	WO-093: ADR for KDBX format-version selection logic
4b93389	WO-091: Plugin signature verification gate with MetadataLoadContext

Built with Cursor and SoftwareForge.ai during the Forge Hackathon 2026 - August 11, 2026.